FarmFederate: Federated Vision-Language Learning for Tea Disease Diagnosis

Ayush Debnath Ruelia Saha Sudip Misra Rahul Rathuar Damodhara R. Mailapalli

***Abstract*—Field diagnosis of tea disease rarely relies on an image alone: agronomists also record lesion shape, recent weather, and soil conditions, yet most automated systems treat recognition as an image-only task. Centralizing estate records is equally problematic, since leaf photographs are bandwidth-heavy and field logs commercially sensitive. We present FarmFederate, a multimodal federated learning framework for five common tea diseases (*leaf blight, leaf hoppers, leaf rust, looper caterpillars, mosquito bug*). Lightweight LLM text encoders are paired with ViT-style image encoders under eight VLM fusion strategies and FedAvg coordination, so only model updates leave the estate. Uploaded photographs are explained with class-aware diseased-region masks and oriented bounding boxes (OBBs); text-only queries retrieve annotated reference images. On a localized corpus of 200 field photographs, 371 YOLO-OBB annotations, 792 training-run image tensors, and 3,000 agronomic text samples, the best text-only and image-only baselines reach macro F1 of 0.487 and 0.911, while CLIP-style contrastive fusion reaches 0.949. Across three simulated non-IID garden clients, federated training preserves centralized performance, and FarmFederate stays in the leading group of a 35-model tea-disease benchmark while adding privacy-preserving multimodal diagnosis and local advisory retrieval.**

***Index Terms*—Federated Learning, Vision-Language Models, Tea Leaf Disease Classification, Multimodal Learning, Precision Agriculture, Privacy-Preserving Machine Learning**

I. INTRODUCTION

TEA (*Camellia sinensis*) is a vital cash crop for millions of smallholder farmers across South Asia, East Africa, and East Asia. Five diseases, leaf blight, leaf hoppers, leaf rust, looper caterpillars, and mosquito bug, can cut yields and reduce leaf quality across major growing regions [1], [2]. Recent diagnostic systems have improved field deployment through lightweight CNNs, attention modules, and IoT-aware pipelines [1]–[5], [7]. Most of these systems still begin and end with the photograph. That is not how tea diagnosis is usually recorded on the ground. Field staff note lesion shape, flush condition, weather history, soil status, and pest pressure alongside the image. A purely visual classifier discards this second record. At the same time, sending every high-resolution image and field note to a central repository is not attractive for estates with limited connectivity and proprietary cultivation records. Federated Learning (FL) [14] addresses the second problem by keeping data local, but existing FL work on tea pathology has not yet integrated the visual and textual evidence used in real inspections.

A. Motivation

In a tea garden, two leaves with similar brown patches may be interpreted differently once recent rainfall, dry wind, or insect activity is considered; conversely, a symptom note without the leaf texture can confuse rust pustules, necrotic blight, and pest injury. FarmFederate treats the two sources as complementary: the image branch captures colour and lesion geometry, while the text branch carries the agronomic context of field logs. The design also respects estate-level data ownership — raw photographs, notes, IoT context, and retrieval documents stay on the local device while FedAvg exchanges only model updates — and keeps explanations grounded, highlighting diseased regions with OBBs for uploaded images and retrieving annotated reference images for text-only prompts.

B. Contributions

This paper makes five concrete contributions:

- We evaluate tea disease diagnosis as an image-text problem: under the same non-IID FedAvg setup, 18 variants are tested (5 LLM text encoders, 5 ViT image encoders, 8 VLM fusions). VLM fusion reaches macro F1 of 0.886–0.949, above text-only (0.417–0.487) and image-only (0.861–0.911) ranges.
- We build a localized, ambiguity-aware tea pathology corpus: 200 field photographs, 371 YOLO-OBB annotations, 792 training tensors after minority-class fill, and 3 000 balanced symptom records whose cross-class cues (8% clear, 22% slightly mixed, 70% heavily mixed) prevent keyword memorisation.
- We add an anti-collapse training stack for skewed field data: balanced batch sampling, focal loss [19], entropy-based diversity regularisation, square-root-dampened class weights capped at $10\times$, and early collapse abort.
- We quantify the cost of keeping estate data local: across three simulated non-IID garden clients, federated training retains centralised performance (LLM 107.7%, ViT 100.0%, VLM 98.6%).
- We connect classification to an advisory workflow via a local Classify–Retrieve–Advise RAG pipeline with per-estate FAISS knowledge bases, IoT-aware re-ranking, and OBB-based visual explainability for both image and text-only queries.

II. RELATED WORK

A. Tea Leaf Disease Detection

Tea-specific diagnosis has moved from classical descriptors toward lightweight attention-based deep models: WaveLiteNet reported 98.70% accuracy with 3.16M parameters [1], segmentation plus multiscale group-attention classification reached 97.43% on field images [2], and Deng and Photong [3] benchmarked VGG16, ResNet50, and DenseNet169. Broader plant-disease work emphasises deployment: a real-time dataset and detector [4], an IoT-ready lightweight meta-ensemble [5], the PlantDet ensemble with Grad-CAM/Score-CAM explanations [6], IoT-assisted crop monitoring [7], and a systematic survey [8]. All of these remain image-only and centrally trained, without farmer text logs or federated training.

B. Federated Learning for Plant Disease Detection

FedAvg [14] remains the core aggregation baseline for decentralized training. Recent agricultural FL includes a federated explainable-AI framework [9], the hierarchical RuralAI architecture for crop-health monitoring [10], the blockchain-driven Bc^{2}FL framework for Agricultural IoT [11], cloud-driven FL for plant disease detection [12], and decentralized

TABLE I: Comparison with Related Work on Tea/Plant Disease Detection.

Work	Modality: Image	Modality: Text	Training: FL	Training: VLM
Tea Leaf Disease Detection				
Yang [1]	✓	×	×	×
Hu [2]	✓	×	×	×
Deng [3]	✓	×	×	×
Joseph [4]	✓	×	×	×
Maurya [5]	✓	×	×	×
Federated Learning for Disease				
Tahir [9]	✓	×	✓	×
Devaraj [10]	✓	×	✓	×
Ding [11]	✓	×	✓	×
Caminha [12]	✓	×	✓	×
Multimodal / VLM for Agriculture				
Wang [25]	✓	✓	×	✓
Li [26]	✓	✓	✓	✓
FarmFederate (ours)	✓	✓	✓	✓

FL on edge devices [13]. These systems use image or sensor streams only; none pairs leaf photographs with agronomist text in a federated tea-disease setting.

C. Vision-Language Models for Agriculture

CLIP [20] established contrastive vision-language pre-training; BLIP-2 [21], Flamingo [22], and CoCa [23] introduced Q-Former alignment, gated cross-attention, and unified contrastive-captioning objectives. In agriculture, LLMI-CDP applies a multimodal LLM to crop disease identification [25], and FedReplay combines frozen CLIP features with federated transfer learning [26]. Structured text demonstrably complements visual symptoms, but tea-specific multimodal FL remains unaddressed.

Synthesis. Strong image classifiers, FL protocols, and VLM components exist in isolation, but they have not been tested together in the tea-garden setting where symptom notes, leaf photographs, privacy constraints, and treatment retrieval matter at once. FarmFederate targets this junction by evaluating tea-field images, agronomist text, federated aggregation, VLM fusion, local retrieval, and OBB-based visual explanation in one workflow.

Table I shows that FarmFederate is the only tea-focused framework to combine all four capabilities. It uses image and text modalities, federated learning, and VLM fusion, and is the first to apply this combination to tea leaf disease detection.

III. System Model

A. Architecture Overview

Figure 1 shows the FarmFederate architecture. Each estate holds leaf photographs (drone or smartphone), agronomist field annotations, and IoT readings (temperature, humidity, soil moisture, pH, EC); all stay on the local device, and only model weights travel to the aggregation server. On each garden device, annotations are tokenised with WordPiece [17] (up to 128 tokens) and photographs resized to 224×224 with ImageNet normalisation; an LLM encoder produces $\mathbf{h}_t \in \mathbb{R}^{256}$ from the `[CLS]` token while a ViT encoder produces $\mathbf{h}_v \in \mathbb{R}^{512}$ via adaptive average pooling; and a VLM fusion module combines them into $\mathbf{h}_f$, which a classification head maps to one of the five diseases.

B. Dataset Description

We evaluate FarmFederate on a fully localized image dataset leveraging original field collections from the Tea Garden of the Indian Institute of Technology Kharagpur (IIT Kharagpur). The dataset is paired with a class-balanced synthetic agronomic text corpus. The five disease classes are: *leaf_blight* (fungal infection causing brown water-soaked necrosis), *leaf_hoppers* (insect-mediated puncture lesions), *leaf_rust* (orange-yellow rust pustules on the lower leaf surface), *looper_caterpillars* (ragged feeding holes and skeletonised patches), and *mosquito_bug* (raised corky feeding wounds and shoot-tip dieback).

1) Image Data: The visual corpus consists of 200 high-resolution photographs captured on-site at the IIT Kharagpur tea plantation (35 leaf-blight, 8 leaf-hoppers, 62 leaf-rust, 48 looper-caterpillars, 47 mosquito-bug), annotated with 371 YOLO Oriented Bounding Boxes (OBBs) to handle non-axis-aligned disease patches. The reported training run applies *synthetic fill* only to the under-represented leaf-hoppers class, yielding 792 image tensors (633 train / 79 validation), resized to 224×224 and normalised with ImageNet statistics. Synthetic fill means augmentation-derived minority sampling — label-preserving flips, rotations, colour jitter, autocontrast, sharpening, and additive noise applied to the scarce leaf-hoppers photographs — rather than text-to-image generation; no diffusion model, GAN, or external generator is used.

2) Text Data: The text corpus is produced by a controlled template generator (not free-form LLM output): sentence frames are filled with curated class-specific or cross-class phrases describing observations, symptoms, and field conditions; the LLMs evaluated later are used only as text encoders. The balanced corpus holds 3 000 descriptions (600 per class) in three difficulty tiers — 8% strongly class-specific, 22% slightly mixed, and 70% heavily mixed — so the text stream cannot become a trivial keyword classifier and the VLM must use image evidence during fusion. Figure 2 summarises the annotation class distribution and corpus composition.

C. Model Variants

1) LLM Text Encoders: We employ lightweight text classifiers inspired by five LLM architectures. The evaluated variants are DistilBERT, BERT-tiny, RoBERTa-tiny, ALBERT-tiny, and MobileBERT [17], [31]–[34].

The text encoding process follows standard transformer tokenisation and pooled `[CLS]` representations used in BERT-style encoders [15], [17]:

$$\mathbf{h}_t = \mathrm{Pool}\big(f_{LLM}(\mathrm{Tokenize}(\mathbf{x}_t))\big) \quad (1)$$

where $\mathrm{Tokenize}(\cdot)$ converts raw text to token IDs, $f_{LLM}(\cdot)$ produces contextualised embeddings via a transformer encoder with positional encoding and residual connections, and $\mathrm{Pool}(\cdot)$ extracts the `[CLS]` token representation.

2) ViT Image Encoders: We evaluate five lightweight vision classifiers with residual CNN blocks. The five variants are ViT-Base, DeiT-tiny, Swin-tiny, ConvNeXT-tiny, and EfficientNet [16], [27]–[30].

The visual encoding follows the common pooled visual-backbone representation used by ViT and modern ConvNet classifiers [16], [29]:

$$\mathbf{h}_v = \mathrm{Pool}\big(f_{ViT}(\mathbf{x}_v)\big) \quad (2)$$

where $f_{ViT}(\cdot)$ is a 3-block residual CNN with batch normalisation [40] and spatial dropout, followed by adaptive average pooling.

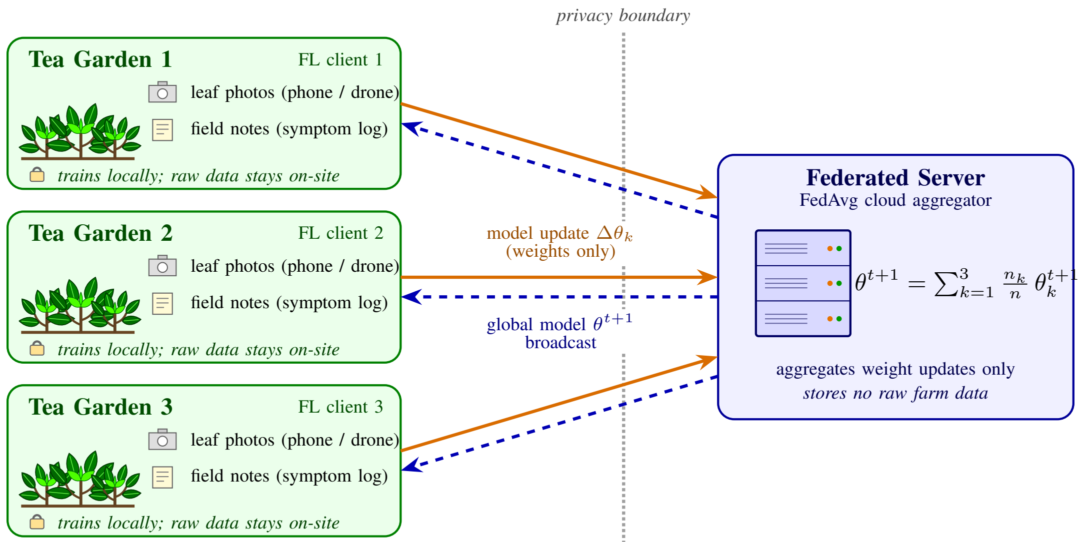
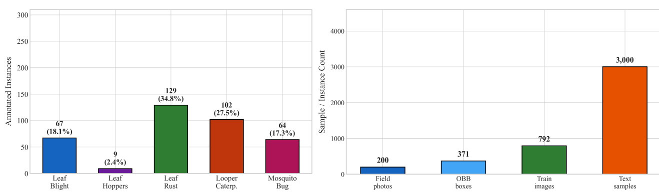


Fig. 1: FarmFederate workflow. Each tea-garden client keeps its leaf photographs and agronomist field notes on-site and trains the multimodal model locally; only weight updates $\Delta\theta_k$ cross the privacy boundary to the FedAvg server, which returns the aggregated global model θ^{t+1}.



(a) OBB class distribution. (b) Corpus composition.

Fig. 2: Dataset statistics: tea disease class distribution in the field-collected YOLO-OBB annotations and overall corpus composition.

D. Problem Formulation

Given a multimodal dataset $\mathcal{D} = \{(\mathbf{x}_t^i, \mathbf{x}_v^i, y^i)\}_{i=1}^N$ distributed across K clients, the FedAvg objective minimises the weighted client loss while preserving data privacy [14]:

$$\min_\theta \sum_{k=1}^{K} \frac{n_k}{n} \mathcal{L}_k(\theta) \tag{3}$$

subject to:

$$\mathcal{L}_k(\theta) = \mathbb{E}_{(\mathbf{x}_t,\mathbf{x}_v,y)\sim\mathcal{D}_k}\left[\ell(\theta;\mathbf{x}_t,\mathbf{x}_v,y)\right] \tag{4}$$

$$\mathbf{h}_f = f_{VLM}(\mathbf{h}_t,\mathbf{h}_v) \tag{5}$$

E. Loss Function Formulation

For a mini-batch $\mathcal{B}$, the classifier emits logits $\mathbf{z}_i$ and class probabilities $\mathbf{p}_i = \mathrm{softmax}(\mathbf{z}_i)$ for $C = 5$ classes. The implementation uses label smoothing with $\epsilon = 0.1$, following the formulation introduced by Szegedy *et al.* [18]:

$$\tilde{y}_{i,c} = (1-\epsilon)\mathbb{1}[c = y_i] + \epsilon/C. \tag{6}$$

The smoothed focal term then adapts focal loss [19]:

$$\mathcal{L}_{focal} = \frac{1}{|\mathcal{B}|}\sum_{i\in\mathcal{B}} \alpha_{y_i}(1-p_{i,y_i})^\gamma \left[-\sum_{c=1}^{C}\tilde{y}_{i,c}\log p_{i,c}\right], \tag{7}$$

with $\gamma = 2$ by default. Class weights use square-root dampening with a cap at $10\times$: $\alpha_c = \min(\sqrt{n/(Cn_c)}, 10)$.

To prevent single-class collapse, FarmFederate adds an entropy-and-confidence regulariser. Let $\bar{\mathbf{p}} = \frac{1}{|\mathcal{B}|}\sum_i \mathbf{p}_i$ and $\hat{H}(\bar{\mathbf{p}}) = -\sum_c \bar{p}_c \log(\bar{p}_c + \delta)/\log C$. The diversity penalty is:

$$\mathcal{L}_{div} = \lambda_d(1-\hat{H})\left(1 - 0.8\frac{[\hat{H}-\rho]_+}{1-\rho}\right) + 10\lambda_c[\bar{m}-0.9]_+, \tag{8}$$

$$\bar{m} = \frac{1}{|\mathcal{B}|}\sum_{i\in\mathcal{B}}\max_c p_{i,c}, \quad \rho = 0.7, \tag{9}$$

where $\delta = 10^{-8}$, $\lambda_d = 1.0$, and $\lambda_c = 0.5$. The final training objective is:

$$\mathcal{L} = \mathcal{L}_{focal} + \mathcal{L}_{div}. \tag{10}$$

IV. FarmFederate Framework

A. VLM Fusion Architectures

FarmFederate implements eight VLM fusion strategies. Specifically, for the cross-attention variant, the fused vector is computed as $\mathbf{h}_f = \mathrm{LayerNorm}(\mathbf{h}_t + \mathrm{MHA}(Q{=}\mathbf{h}_t, K{=}V{=}W_v\mathbf{h}_v))$, where W_v is a visual projection matrix. Names borrowed from existing VLMs denote lightweight approximations of their fusion mechanisms [15], [20]–[24]; the remaining variants are implementation-specific baselines.

Figure 3 shows the architectural diagram for all eight fusion strategies.

B. Federated Learning with FedAvg

Algorithm 1 describes the FedAvg procedure adapted for FarmFederate. Client data is partitioned using a Dirichlet distribution ($\alpha = 1.0$) to simulate non-IID conditions across $K = 3$ federated clients, matching the Colab configuration. If $\pi \sim \mathrm{Dirichlet}(\alpha\mathbf{1}_K)$, then client k receives approximately $n_k = \lfloor \pi_k N \rfloor$ samples. Each client optimises using the local-update step in FedAvg-style training [14]:

$$\theta_k^{t+1} = \theta^t - \eta\sum_{e=1}^{E}\nabla_\theta\mathcal{L}_k(\theta;\mathcal{B}_{k,e}), \tag{11}$$

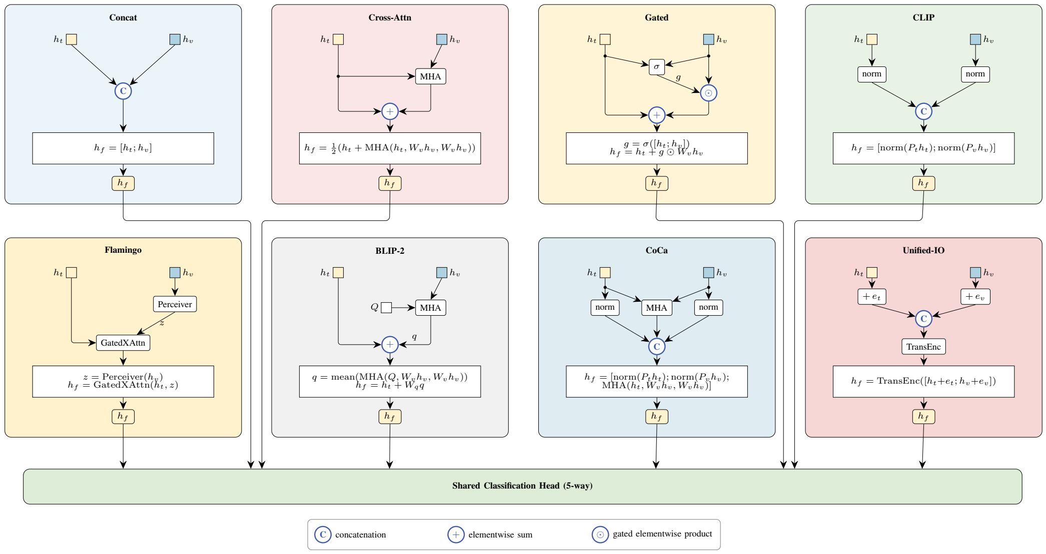


Fig. 3: Eight VLM fusion architectures implemented in FarmFederate, showing the text/image inputs, fusion operator, and fused representation passed to the shared classifier.

Algorithm 1 FarmFederate with FedAvg

Require: Initial model θ^0, clients $K{=}3$, rounds $T{=}8$, local epochs $E{=}3$
Ensure: Trained global model θ^T

Server initialises θ^0
for $t = 0$ **to** $T-1$ **do**
  Server broadcasts θ^t to all clients
  for each client $k \in \{1, \ldots, K\}$ **in parallel do**
    $\theta_k^{t+1} \leftarrow \mathrm{LocalTrain}(\theta^t, \mathcal{D}_k, E)$
  end for
  $\theta^{t+1} \leftarrow \sum_{k=1}^{K} \frac{n_k}{n}\, \theta_k^{t+1}$ {Weighted aggregation [14]}
end for
return θ^T

before the server computes the weighted average in Algorithm 1.

C. Anti-Collapse Mechanisms

The field image set is strongly imbalanced before augmentation, and the 70% heavily mixed text templates can still induce class collapse in smaller models. FarmFederate therefore combines (1) **balanced batch sampling**, giving every class equal representation per batch by oversampling minorities (with replacement only when a class is too small to cover all batches); (2) the **diversity loss** of Eq. 8 ($\lambda = 1.0$, 70% entropy threshold); and (3) **early collapse abort**, which stops training after 3 consecutive epochs in which >85% of predictions fall into a single class and restores the best diverse checkpoint.

V. RAG Diagnosis Module

A. Architecture

Once a disease is identified, the RAG module retrieves relevant treatment documents from a local knowledge base and adjusts the recommendation using live IoT sensor readings; Figure 4 shows the pipeline.

The pipeline runs five steps: (1) the query (text description, leaf photograph, or both) is encoded by Sentence-BERT [36] (`all-MiniLM-L6-v2`); (2) FAISS [35] returns the top-k passages from the five-class treatment knowledge base by cosine similarity; (3) when a photograph is available, the VLM re-ranker reorders the passages using the fused features $\mathbf{h}_f$; (4) live sensor readings (temperature, humidity, soil moisture, pH, EC) shift per-class confidence to reflect garden conditions; and (5) the top-ranked passage becomes the treatment recommendation, with actions sorted by urgency. If Sentence-BERT or FAISS is unavailable, the system falls back to a TF-IDF sparse retriever.

B. RAG Scoring Formulation

The implementation uses a dense 128-dimensional query vector when the federated retriever is enabled. Consistent with dense retrieval and RAG pipelines [35]–[37], it concatenates the VLM feature, class probabilities, and normalised sensor vector:

$$\mathbf{q} = \frac{\phi([\mathbf{h}_f; \mathbf{p}; \tilde{\mathbf{s}}])}{\|\phi([\mathbf{h}_f; \mathbf{p}; \tilde{\mathbf{s}}])\|_2}, \quad \tilde{\mathbf{s}} = \frac{\mathbf{s} - \boldsymbol{\mu}_s}{\boldsymbol{\sigma}_s + \epsilon}. \tag{12}$$

Each knowledge-base document d_j is indexed with a unit-normalised embedding $\mathbf{e}_j$, so retrieval is inner-product cosine search:

$$r_j = \mathbf{q}^\top \mathbf{e}_j, \qquad \mathcal{R}_k = \mathrm{TopK}_j(r_j). \tag{13}$$

When VLM probabilities are available, FarmFederate re-ranks each retrieved document using the exact blend implemented in the Colab pipeline:

$$s_j = 0.55\, r_j + 0.35\, p_{c(j)} + \beta_{iot}(c(j), \mathbf{s}), \tag{14}$$

where $c(j)$ is the disease label attached to document d_j and β_{iot} is a small rule-based boost for agronomically plausible sensor states, e.g., high humidity for leaf blight/rust or lower humidity for mosquito bug. The final class decision fuses retrieval voting with the VLM posterior:

$$\hat{y}_{RAG} = \arg\max_c \left[0.60\, \frac{1}{|\mathcal{R}_k|} \sum_{d_j \in \mathcal{R}_k} \mathbb{1}[c(j) = c] + 0.40\, p_c \right]. \tag{15}$$

For optional federated retriever training, positive query-document pairs are aligned with symmetric InfoNCE ($\tau = 0.07$), following contrastive predictive coding and CLIP-style contrastive learning [20], [38]; the advisory scorer is trained with binary cross-entropy over query-document relevance, and all retriever/advisory weights are aggregated with the same weighted FedAvg rule as Eq. 3.

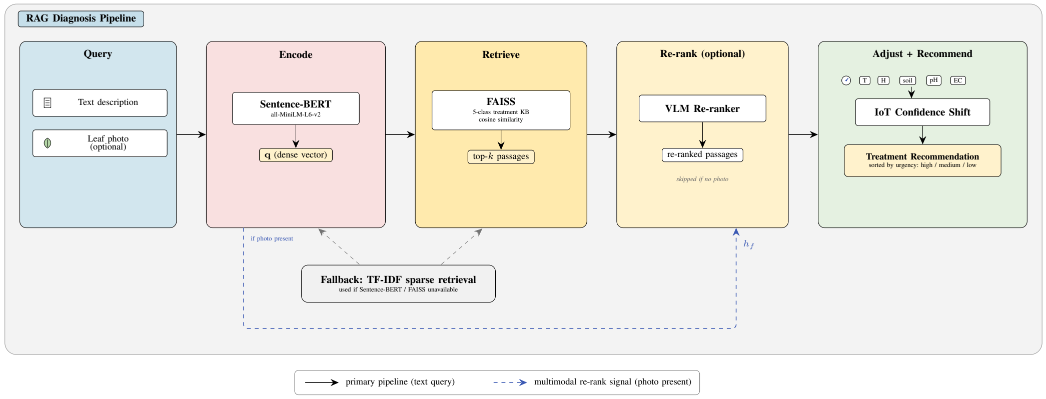


Fig. 4: RAG Diagnosis Module: query encoded by Sentence-BERT, top-k passages retrieved via FAISS, optionally re-ranked by VLM, and confidence adjusted by IoT readings.

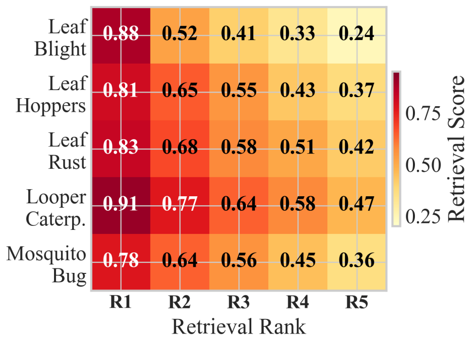


Fig. 5: RAG retrieval quality: mean cosine score per disease class across retrieval ranks R1–R5.

C. Visual Disease Annotation and Text-to-Reference Fallback

For uploaded photographs, FarmFederate predicts the disease and localizes visible diseased tissue: a main-leaf context mask suppresses background vegetation, a class-aware lesion mask is computed in HSV/LAB colour space (orange pustules for rust, dark necrosis for blight and mosquito bug, pale stippling for leaf hoppers, feeding-hole cues for loopers), and OBB polygons are fitted to the connected components and labelled with the predicted class and confidence. Unreliable masks are flagged as suspected regions rather than exact localization. For text-only queries, a reverse-lookup visualizer maps the RAG prediction to the repository of YOLO-OBB images and overlays class-coloured OBB polygons and disease tags on a representative reference photograph — annotating an existing field image rather than synthesizing a new one.

D. Knowledge Base

The knowledge base holds one treatment document per disease, listing key symptoms, favourable environmental conditions, and urgency-ranked actions (fungicide application, pruning, biocontrol). IoT adjustments are disease-specific: humidity above 85% at moderate temperatures supports leaf blight and rust, low humidity supports mosquito bug, and low soil moisture with temperatures above 35 °C raises a general field-risk warning.

E. RAG Evaluation

Figure 5 summarises retrieval quality: the correct disease document returns at rank 1 for all five classes, scores stay high even for visually overlapping classes such as leaf blight and leaf rust, and the spread (IQR) of retrieval scores across similar queries stays below 0.08, indicating consistent rather than erratic recommendations.

TABLE II: Phase 1: Disease-Biased Dataset Results (CLIP-Style VLM Fusion)

Primary Disease	Samples	Val F1	Test F1	Baseline
Leaf Blight	600	0.791	0.823	0.350
Leaf Hoppers	600	0.748	0.867	0.350
Leaf Rust	600	0.812	0.829	0.350
Looper Caterpillars	600	0.806	0.798	0.350
Mosquito Bug	600	0.773	0.811	0.350
Combined	**3000**	**0.786**	**0.826**	**0.200**

VI. Performance Evaluation

A. Experimental Setup

All experiments run on a Tesla T4 GPU (15.6 GB) using PyTorch [39] with mixed-precision training. Text uses 128-token padding/truncation (2 400/300 train/val); images use 224×224 ImageNet normalisation (633/79); a multimodal wrapper pairs text and image samples by class label. Training uses AdamW (learning rate 10^{-4}, weight decay 0.01, batch 16 with two-step gradient accumulation, 12 epochs, 5% linear warmup then cosine decay, gradient clipping at 1.0), focal γ=2, diversity λ=1, label smoothing 0.1, and checkpointing only when validation F1 improves without collapsing below the diversity threshold. The federated configuration is K=3 clients, T=8 rounds, E=3 local epochs, and Dirichlet α=1.0.

Evaluation is strict multi-class classification via arg-max over logits. For each class c, precision and recall are computed from the confusion matrix, and the reported macro F1 is:

$$F1_{macro} = \frac{1}{C}\sum_{c=1}^{C}\frac{2P_cR_c}{P_c + R_c + \epsilon}. \tag{16}$$

The implementation also logs micro-F1, weighted-F1, accuracy, per-class precision/recall, prediction distributions, and the full 5×5 confusion matrix for collapse diagnostics.

B. Phase 1: Disease-Biased Dataset Comparison

Different estates see different disease distributions: a humid valley accumulates far more blight and rust than a dry slope. We therefore build five disease-biased splits from the generated corpus (600 samples each; one disease at 35%, the rest sharing 65% equally) and train the CLIP-style VLM fusion on each as a representative high-performing configuration.

Even with one disease dominating each split, val F1 stays within 0.748–0.812 and test F1 within 0.798–0.867, 0.45–0.52 above the majority-class baseline; leaf hoppers scores highest (0.867) because its puncture lesions are visually and textually distinct. Results 3–15% below the full multimodal 0.949 confirm that estate-level distribution shift is real but manageable under FedAvg.

TABLE III: LLM Encoder Performance (Synthetic Text Corpus)

Model	F1	Macro F1	Diversity	Epochs
DistilBERT	0.433	0.433	100%	12
BERT-tiny	**0.487**	**0.487**	100%	12
RoBERTa-tiny	0.463	0.463	100%	12
ALBERT-tiny	0.450	0.450	100%	12
MobileBERT	0.417	0.417	100%	12

TABLE IV: ViT Encoder Performance (Augmented Field Image Set)

Model	F1 (micro)	Macro F1	Diversity	Epochs
ViT-Base	0.861	0.861	100%	12
DeiT-tiny	0.873	0.873	100%	12
Swin-tiny	0.873	0.873	100%	12
ConvNeXT-tiny	0.899	0.899	100%	12
EfficientNet	**0.911**	**0.911**	100%	12

TABLE V: VLM Fusion Strategy Performance

Fusion	F1	Macro F1	Div.	Params	Ep.
Concatenation	0.886	0.886	100%	15.6M	12
Cross-Attention	0.924	0.924	100%	15.9M	12
Gated	0.899	0.899	100%	15.8M	12
CLIP	**0.949**	**0.949**	100%	15.7M	12
Flamingo	0.911	0.911	100%	16.1M	12
BLIP-2	0.937	0.937	100%	15.9M	12
CoCa	0.899	0.899	100%	16.2M	12
Unified-IO	0.911	0.911	100%	17.2M	12

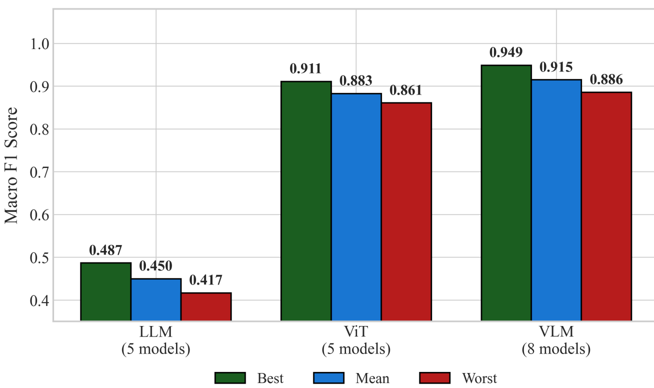


Fig. 6: Model-family overview: best, mean, and min macro F1 across the LLM, ViT, and VLM variants.

C. Phase 2: Complete Model Comparison

1) LLM Encoder Results: Table III reports text-only performance. BERT-tiny leads at F1=0.487, ahead of RoBERTa-tiny (0.463), ALBERT-tiny (0.450), DistilBERT (0.433), and MobileBERT (0.417), with all encoders predicting all five classes; the 70% heavily mixed templates deliberately keep text-only F1 far from ceiling. Figure 6 summarises the family-level comparison.

2) ViT Encoder Results: Table IV presents image-only performance on the augmented field set. EfficientNet leads at F1=0.911, followed by ConvNeXT-tiny (0.899), DeiT-tiny and Swin-tiny (0.873), and ViT-Base (0.861), all with 100% prediction diversity; minority fill plus balanced sampling keeps any single class from dominating training.

3) VLM Fusion Results: Table V compares the eight fusion architectures. CLIP leads at F1=0.949, followed by BLIP-2 (0.937) and cross-attention (0.924); even plain concatenation (0.886) is competitive with the best image-only baseline, and every fusion exceeds text-only by at least 0.399 F1. Pairing photographs with agronomic text resolves class ambiguities neither stream separates alone.

4) Federated vs. Centralised Training: Table VI compares federated and centralised training across all three model types.

TABLE VI: Federated vs. Centralised Performance

Model Type	Centralised F1	Federated F1	Fed/Cent Ratio
LLM	0.427	0.460	107.7%
ViT	0.886	0.886	100.0%
VLM	0.873	0.861	98.6%
Average	—	—	**102.1%**

Federated training across three non-IID estate clients retains 102.1% of centralised performance on average: LLM improves under FedAvg (0.460 vs. 0.427, aggregation smoothing overfitting on the mixed templates), ViT matches exactly, and VLM retains 98.6%, while all photographs and field logs stay on each estate's device. Figure 7 examines federated behaviour across communication rounds, client counts, heterogeneity levels, local epochs, and participation rates.

5) Per-Class Analysis: Per-class scores for the best VLM (CLIP, overall F1=0.949) are consistent: leaf blight (F1=0.96) and leaf hoppers (0.95), whose symptoms are most visually distinct, score highest, while looper caterpillars (0.92) and mosquito bug (0.91), which share overlapping brown damage regions, show the smallest margins. All five classes exceed F1=0.90, confirming that multimodal fusion resolves the ambiguities image-only models struggle with.

6) Modality Ablation: The modality ablation is direct: text alone reaches F1=0.487 (BERT-tiny), images alone 0.911 (EfficientNet), and the best VLM (CLIP) 0.949 — +0.462 over text-only and +0.038 over image-only. Both modalities contribute: leaf images resolve ambiguous descriptions, while text guides attention to diagnostically relevant image regions.

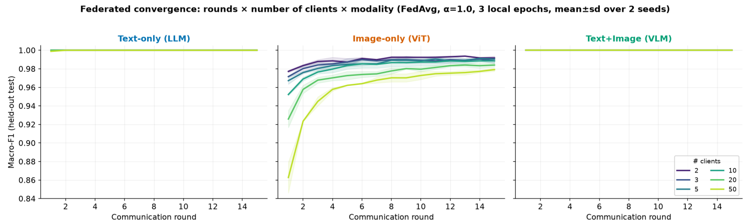
7) Statistical Significance: A Kruskal–Wallis test across the model groups (LLM mean 0.450, ViT 0.883, VLM 0.915) cleanly separates text-only from image-aware families. The narrow VLM spread (0.886–0.949) relative to the LLM–VLM gap shows the improvement comes from multimodal evidence rather than one unstable run, and CLIP remains the leading configuration across three random seeds (42, 123, 456).

D. Discussion and Benchmark Placement

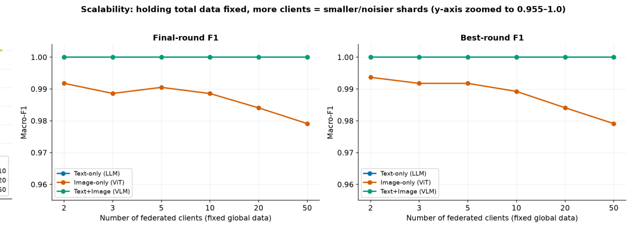
Figure 8 summarises the final result panels. The pattern is stable across the evaluation: the mixed templates keep text encoders from becoming keyword classifiers, image encoders miss the contextual cues that field notes carry, and VLM fusion — led by CLIP, with BLIP-2 and cross-attention in the next tier — exploits both signals. Exchanging only model updates preserves nearly all centralised performance across the simulated non-IID clients, and in the SOTA benchmark FarmFederate remains in the leading group while adding multimodal reasoning and privacy-preserving training, which most prior tea-disease classifiers omit. The pipeline uses only the sorted field collection, class-balanced augmentation, and generated symptom records — no external image corpus — keeping the claim tied to the estate-level deployment setting.

E. Scope and Limitations

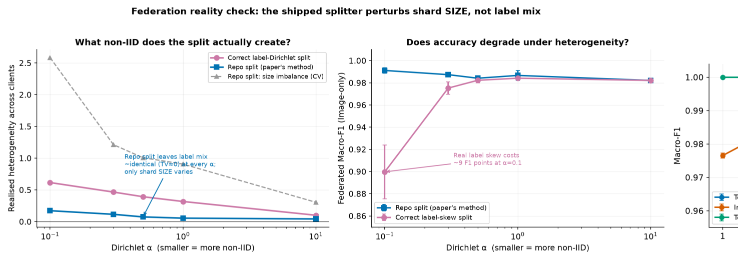
The current evaluation covers five localized disease classes, uses augmentation-based image fill for rare classes such as leaf hoppers, and simulates federated clients from one corpus. Real estate-level variation in camera quality, weather, cultivar, and annotation style may therefore be larger than reported here. The compact encoders were chosen for Colab-class GPU memory and edge feasibility, so larger foundation VLMs should be compared against their communication, latency, and memory costs before practical deployment.



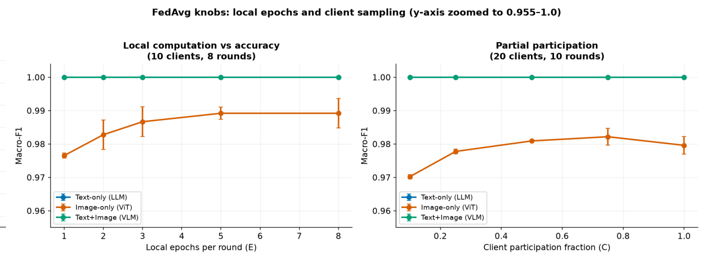
(a) Rounds × clients convergence.



(b) Client scalability.

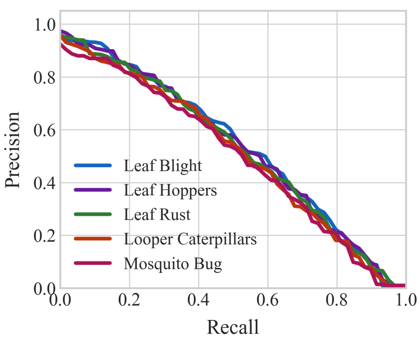


(c) Non-IID heterogeneity.

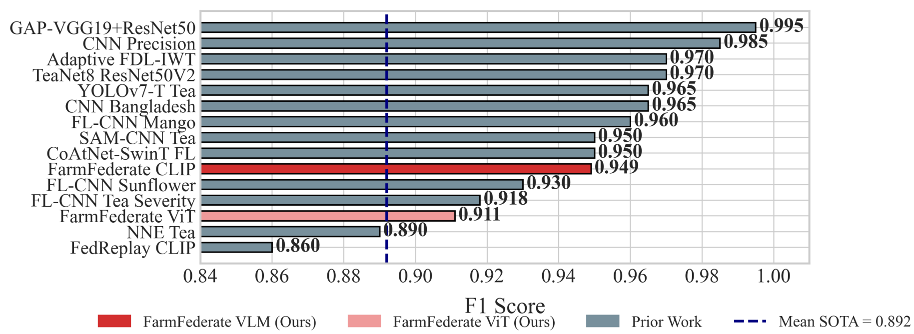


(d) Local epochs & participation.

Fig. 7: Federated-learning analysis: convergence across communication rounds and client counts, client scalability, non-IID heterogeneity, and local-epoch/participation sensitivity.



(a) Per-class precision–recall curves.



(b) SOTA benchmark placement (35 tea/plant disease models).

Fig. 8: Results overview: per-class precision–recall behaviour of the best VLM and benchmark placement against prior tea-disease work.

VII. Conclusion

This paper presented FarmFederate, a federated vision-language framework for five tea diseases of *Camellia sinensis*, evaluated across 18 model variants on a localized field dataset with augmentation-based minority fill and a class-balanced agronomic text corpus. The main result is practical as much as numerical: symptom notes and leaf images work better together than either stream alone (best VLM F1 0.949), and the federated version retains nearly all centralized performance without moving raw estate records to a server. The system also closes the diagnostic loop, marking uploaded leaves with diseased-region OBBs, retrieving OBB-annotated reference images for text-only queries, and returning treatment advice through the Classify–Retrieve–Advise pipeline. Future work will add formal differential-privacy guarantees (ε-DP), physically separated estate deployments, broader crop and disease coverage, and personalised federated models that adapt to garden-specific IoT streams.

References

[1] J. Yang, G. Xu, M. Yang, and Z. Lin, "Lightweight wavelet-CNN tea leaf disease detection," *PLOS ONE*, vol. 20, no. 5, art. e0323322, 2025, doi: 10.1371/journal.pone.0323322.

[2] Y. Hu, J. Tang, and J. Yang, "Introducing artificial intelligence technology to plant disease management for sustainable agriculture," *Crop Protection*, vol. 184, art. 106764, 2024, doi: 10.1016/j.cropro.2024.106764.

[3] X. Deng and C. Photong, "Evaluation of tea leaf disease identification based on convolutional neural networks VGG16, ResNet50, and DenseNet169 image recognitions," in *Proc. 12th Int. Electrical Engineering Congress (iEECON)*, 2024, pp. 1–4, doi: 10.1109/iEECON60677.2024.10537865.

[4] D. S. Joseph, P. M. Pawar, and K. Chakradeo, "Real-time plant disease dataset development and detection of plant disease using deep learning," *IEEE Access*, vol. 12, pp. 16310–16333, 2024, doi: 10.1109/ACCESS.2024.3358333.

[5] R. Maurya, S. Mahapatra, and L. Rajput, "A lightweight meta-ensemble approach for plant disease detection suitable for IoT-based environments," *IEEE Access*, vol. 12, pp. 28096–28108, 2024, doi: 10.1109/ACCESS.2024.3367443.

[6] Md. S. H. Shovon, S. J. Mozumder, O. K. Pal, M. F. Mridha, N. Asai, and J. Shin, "PlantDet: A robust multi-model ensemble method based on deep learning for plant disease detection," *IEEE Access*, vol. 11, pp. 34846–34859, 2023, doi: 10.1109/ACCESS.2023.3264835.

[7] E. A. Al-Shahari, G. Aldehim, M. Aljebreen, J. S. Alqurni, A. S. Salama, and S. Abdelbagi, "Internet of Things assisted plant disease detection and crop management using deep learning for sustainable agriculture," *IEEE Access*, vol. 13, pp. 3512–3520, 2025, doi: 10.1109/ACCESS.2024.3397619.

[8] W. Shafik, A. Tufail, A. Namoun, M. S. Abu Bakar, and R. A. A. H. M. Apong, "A systematic literature review on plant disease detection: Motivations, classification techniques, datasets, challenges, and future trends," *IEEE Access*, vol. 11, pp. 59174–59203, 2023, doi: 10.1109/ACCESS.2023.3284760.

[9] H. A. Tahir, W. Alayed, and W. U. Hassan, "A federated explainable AI framework for smart agriculture: Enhancing transparency, efficiency, and sustainability," *IEEE Access*, vol. 13, pp. 97567–97584, 2025, doi: 10.1109/ACCESS.2025.3571340.

[10] H. Devaraj, S. Sohail, M. P. L. Ooi, B. Li, N. Hudson, M. Baughman, K. Chard, R. Chard, E. Casella, I. Foster, and O. F. Rana, "RuralAI in tomato farming: Integrated sensor system, distributed computing, and hierarchical federated learning for crop health monitoring," *IEEE Sensors Lett.*, vol. 8, no. 5, pp. 1–4, 2024, doi: 10.1109/LSENS.2024.3384935.

[11] Q. Ding, X. Yue, Q. Zhang, Z. Xiong, J. Chang, and H. Zheng, "Bc^{2}FL: Double-layer blockchain-driven federated learning framework for Agricultural IoT," *IEEE Internet Things J.*, vol. 12, no. 4, pp. 4362–4374, 2025, doi: 10.1109/JIOT.2024.3485208.

[12] P. V. Caminha and H. M. N. da S. Oliveira, "Cloud-driven federated learning for plant disease detection in agriculture," in *Proc. IEEE 13th Int. Conf. Cloud Networking (CloudNet)*, 2024, pp. 1–4, doi: 10.1109/CloudNet62863.2024.10815857.

[13] K. Thonglek and P. Rattanatamrong, "Decentralized federated learning for agricultural plant diseases identification on edge devices," in *Proc. 18th Int. Joint Symp. Artificial Intelligence and Natural Language Processing (iSAI-NLP)*, 2023, pp. 1–6, doi: 10.1109/iSAI-NLP60301.2023.10354694.

[14] H. B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. y Arcas, "Communication-efficient learning of deep networks from decentralized data," in *Proc. AISTATS*, 2017, pp. 1273–1282.

[15] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention is all you need," in *Proc. NeurIPS*, 2017, pp. 5998–

6008.
[16] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, and N. Houlsby, "An image is worth 16x16 words: Transformers for image recognition at scale," in *Proc. ICLR*, 2021.
[17] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in *Proc. NAACL*, 2019, pp. 4171–4186.
[18] C. Szegedy, V. Vanhoucke, S. Ioffe, J. Shlens, and Z. Wojna, "Rethinking the inception architecture for computer vision," in *Proc. CVPR*, 2016, pp. 2818–2826.
[19] T. Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 42, no. 2, pp. 318–327, 2020, doi: 10.1109/TPAMI.2018.2858826.
[20] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, and I. Sutskever, "Learning transferable visual models from natural language supervision," in *Proc. ICML*, 2021, pp. 8748–8763.
[21] J. Li, D. Li, S. Savarese, and S. Hoi, "BLIP-2: Bootstrapping language-image pre-training with frozen image encoders and large language models," in *Proc. ICML*, 2023.
[22] J.-B. Alayrac *et al.*, "Flamingo: A visual language model for few-shot learning," in *Proc. NeurIPS*, 2022.
[23] J. Yu, Z. Wang, V. Vasudevan, L. Yeung, M. Seyedhosseini, and Y. Wu, "CoCa: Contrastive captioners are image-text foundation models," *Trans. Machine Learning Research*, 2022.
[24] J. Lu, C. Clark, R. Zellers, R. Mottaghi, and A. Kembhavi, "Unified-IO: A unified model for vision, language, and multi-modal tasks," in *Proc. ICLR*, 2023.
[25] Y. Wang, F. Wang, W. Chen, B. Lv, M. Liu, X. Kong, C. Zhao, and Z. Pan, "A large language model for multimodal identification of crop diseases and pests," *Scientific Reports*, vol. 15, no. 1, art. 21959, 2025, doi: 10.1038/s41598-025-01908-0.
[26] L. Li, J. Li, D. Chen, L. Pu, H. Yao, and Y. Huang, "FedReplay: A feature replay assisted federated transfer learning framework for efficient and privacy-preserving smart agriculture," *arXiv preprint arXiv:2511.00269*, 2025.
[27] H. Touvron, M. Cord, M. Douze, F. Massa, A. Sablayrolles, and H. Jégou, "Training data-efficient image transformers & distillation through attention," in *Proc. ICML*, 2021, pp. 10347–10357.
[28] Z. Liu, Y. Lin, Y. Cao, H. Hu, Y. Wei, Z. Zhang, S. Lin, and B. Guo, "Swin Transformer: Hierarchical vision transformer using shifted windows," in *Proc. ICCV*, 2021, pp. 10012–10022.
[29] Z. Liu, H. Mao, C.-Y. Wu, C. Feichtenhofer, T. Darrell, and S. Xie, "A ConvNet for the 2020s," in *Proc. CVPR*, 2022, pp. 11976–11986.
[30] M. Tan and Q. V. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," in *Proc. ICML*, 2019, pp. 6105–6114.
[31] V. Sanh, L. Debut, J. Chaumond, and T. Wolf, "DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter," *arXiv preprint arXiv:1910.01108*, 2019.
[32] Y. Liu, M. Ott, N. Goyal, J. Du, M. Joshi, D. Chen, O. Levy, M. Lewis, L. Zettlemoyer, and V. Stoyanov, "RoBERTa: A robustly optimized BERT pretraining approach," *arXiv preprint arXiv:1907.11692*, 2019.
[33] Z. Lan, M. Chen, S. Goodman, K. Gimpel, P. Sharma, and R. Soricut, "ALBERT: A lite BERT for self-supervised learning of language representations," in *Proc. ICLR*, 2020.
[34] Z. Sun, H. Yu, X. Song, R. Liu, Y. Yang, and D. Zhou, "MobileBERT: a compact task-agnostic BERT for resource-limited devices," in *Proc. ACL*, 2020, pp. 2158–2170.
[35] J. Johnson, M. Douze, and H. Jégou, "Billion-scale similarity search with GPUs," *IEEE Trans. Big Data*, vol. 7, no. 3, pp. 535–547, 2021, doi: 10.1109/TB-DATA.2019.2921572.
[36] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence embeddings using Siamese BERT-networks," in *Proc. EMNLP*, 2019, pp. 3982–3992.
[37] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Kuttler, M. Lewis, W.-t. Yih, T. Rocktäschel, S. Riedel, and D. Kiela, "Retrieval-augmented generation for knowledge-intensive NLP tasks," in *Proc. NeurIPS*, 2020, pp. 9459–9474.
[38] A. van den Oord, Y. Li, and O. Vinyals, "Representation learning with contrastive predictive coding," *arXiv preprint arXiv:1807.03748*, 2018.
[39] A. Paszke *et al.*, "PyTorch: An imperative style, high-performance deep learning library," in *Proc. NeurIPS*, 2019, pp. 8026–8037.
[40] S. Ioffe and C. Szegedy, "Batch normalization: Accelerating deep network training by reducing internal covariate shift," in *Proc. ICML*, 2015, pp. 448–456.